

COS 433: Cryptography

Reminders:

- Problem Set 1 due today!
- Problem Set 2 to be released today

Last time: CPA Security

IND-CPA Security

Challenger/Honest Party



Sampled once

k

$b \leftarrow \{0,1\}$

$m_0^{(\lambda)}, m_1^{(\lambda)}$

$\text{Enc}(k, m_b; r)$

New Enc randomness
each time

Adversary



b'

Win if $b' = b$

Definition: A secret-key encryption scheme with message length $n = n(\lambda)$ satisfies **chosen plaintext attack (IND-CPA) security** if **for all** $t(\lambda) = \text{poly}(\lambda)$,

For all polynomial-time



$$\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$$

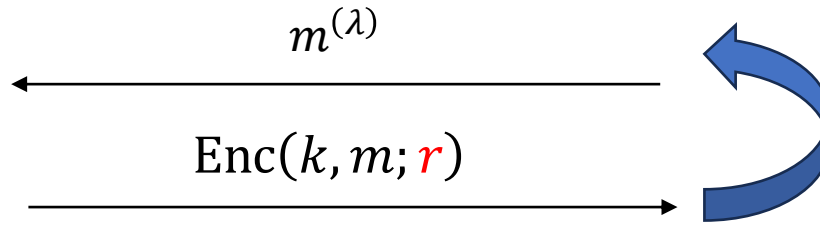
IND-CPA Security, Alt. Definition

Challenger/Honest Party



k

$b \leftarrow \{0,1\}$

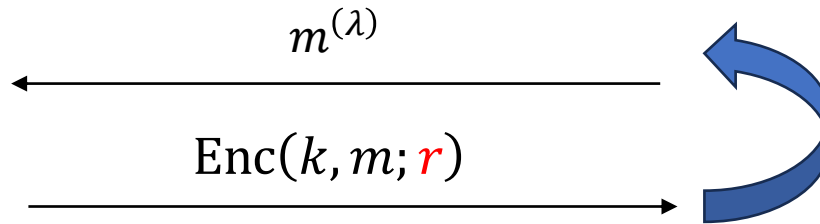
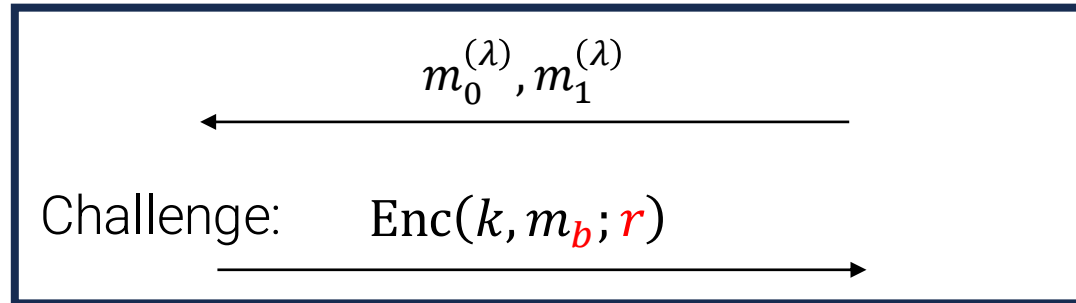


Adversary



b'

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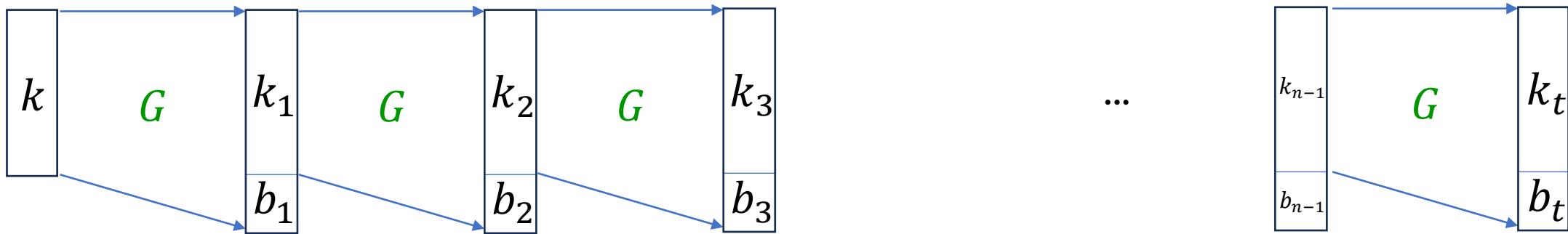
For all polynomial-time



$$\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$$

Building CPA-secure Encryption

Partial Solution: Stream Cipher



Idea: To encrypt the i^{th} message, use b_i as a one-time pad (PRG length extension)

Secure! (won't prove)

But: parties must keep a counter for how many messages sent so far (and remember k_t).

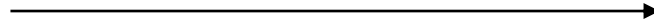
“Stateful encryption” has many drawbacks (and does not satisfy our definition).

Building CPA-secure Encryption



k, m

$$ct = G(k) \oplus m$$



k

$$m' = G(k) \oplus ct$$

$$G: \{0,1\}^\lambda \rightarrow \{0,1\}^n$$

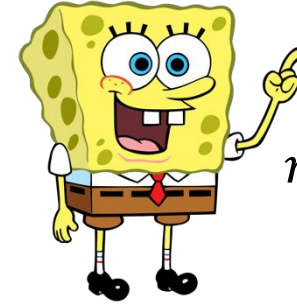
- We need to make use of **randomness**
- Problem: Bob doesn't get access to Alice's random coins... how to decrypt?
- **Idea:** can Alice make the coins **public**?

Building CPA-secure Encryption



k, m, r

$$ct = (r, F(k, r) \oplus m)$$



k

$$m' = F(k, ct_1) \oplus ct_2$$

$$F: \{0,1\}^\lambda \times \{0,1\}^\lambda \rightarrow \{0,1\}^n$$

What property of F would guarantee CPA-security?

Pseudorandom Functions (PRFs)

Challenger



$b \leftarrow \{0,1\}$

$b = 0: k \leftarrow \{0,1\}^\lambda$

$b = 1: F \leftarrow \text{Fun}(\{0,1\}^n \rightarrow \{0,1\}^m)$

$x \in \{0,1\}^n$

$b = 0: \text{PRF}(k, x)$

$b = 1: F(x)$

Adversary



b'

Win if $b' = b$

Definition: An efficiently computable function family $\text{PRF}: \{0,1\}^\lambda \times \{0,1\}^n \rightarrow \{0,1\}^m$ is a PRF family if:

For all polynomial-time

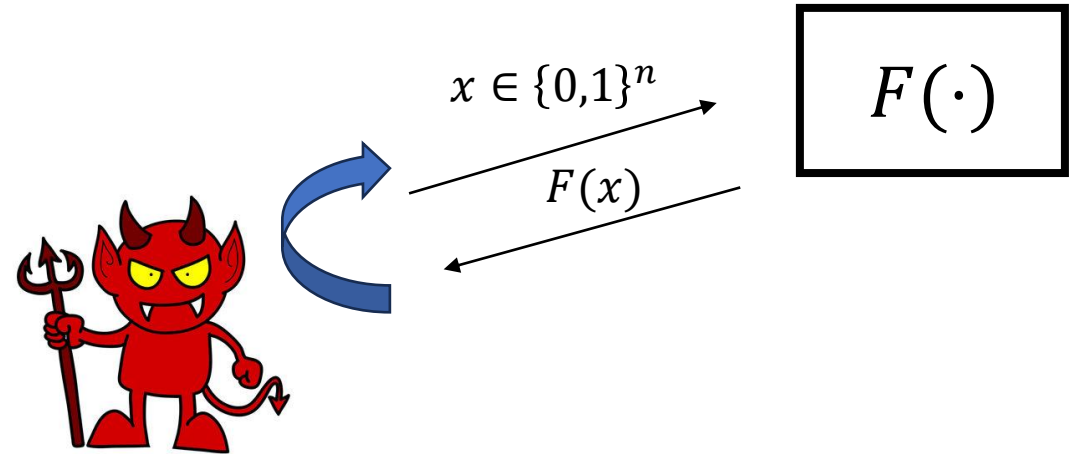


$$\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$$

Pseudorandom Functions (PRFs)

Another view: “Oracle algorithms”

A poly-time **oracle algorithm** $A^{F(\cdot)}(1^\lambda)$ is an algorithm that can call a function $F(\cdot)$ on inputs of length $\text{poly}(\lambda)$ **at unit cost**



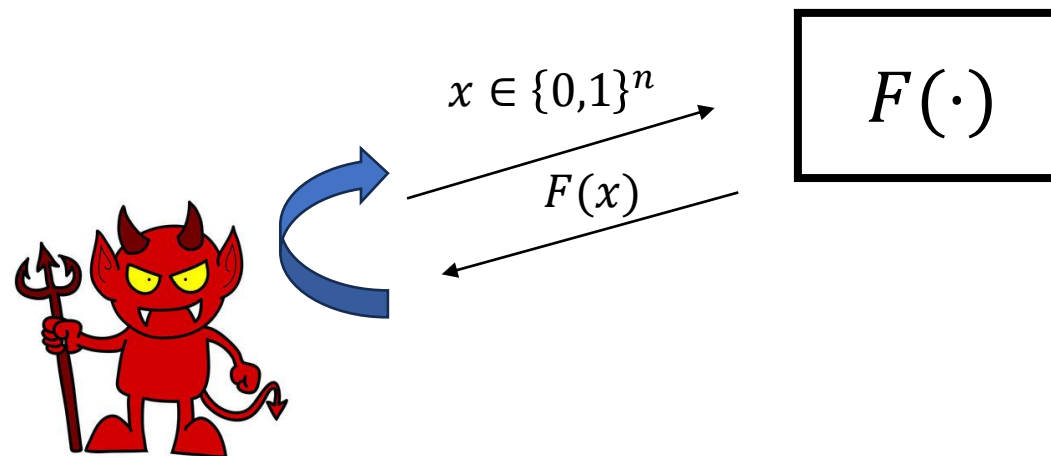
Definition: An efficiently computable function family $\text{PRF}: \{0,1\}^\lambda \times \{0,1\}^n \rightarrow \{0,1\}^m$ is a PRF family if:

$$\text{For all polynomial-time } A^{F(\cdot)}, A^{\text{PRF}(\textcolor{teal}{k}, \cdot)}(1^\lambda) \approx_c A^{\textcolor{teal}{F}(\cdot)}(1^\lambda)$$

Pseudorandom Functions (PRFs)

Another view: “Oracle algorithms”

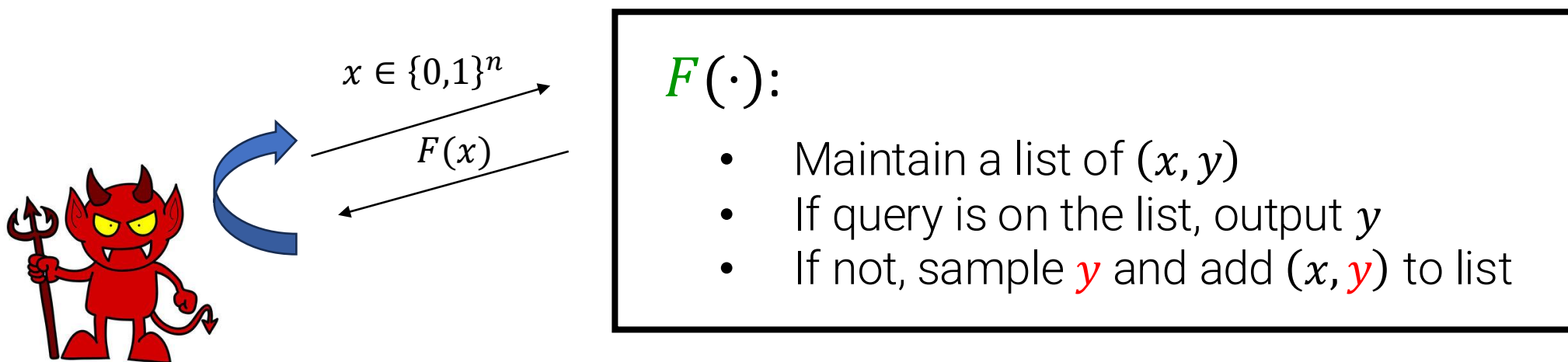
A poly-time **oracle algorithm** $A^{F(\cdot)}(1^\lambda)$ is an algorithm that can call a function $F(\cdot)$ on inputs of length $\text{poly}(\lambda)$ **at unit cost**



Aside: can we efficiently compute $A^{F(\cdot)}(1^\lambda)$? A uniformly random $F(\cdot)$ is likely to be extremely hard to compute.

Observation: we can efficiently evaluate $A^{F(\cdot)}(1^\lambda)$ with a **stateful algorithm**

Pseudorandom Functions (PRFs)



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Adversary



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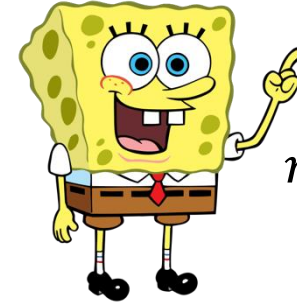
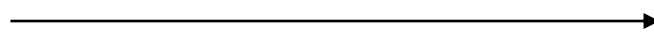
$$\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$$

Building CPA-secure Encryption



k, m, r

$$ct = (r, \text{PRF}(k, r) \oplus m)$$



k

$$m' = F(k, ct_1) \oplus ct_2$$

$$\text{PRF}: \{0,1\}^\lambda \times \{0,1\}^\lambda \rightarrow \{0,1\}^n$$

Theorem: if PRF is a **PRF family**, then this encryption scheme is **CPA-secure**!

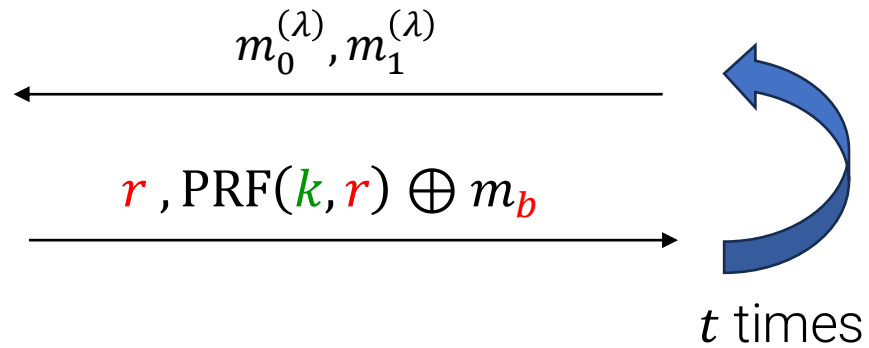
Proof: Hybrid argument & security reduction!

Building CPA-secure Encryption



k

$b \leftarrow \{0,1\}$



b'

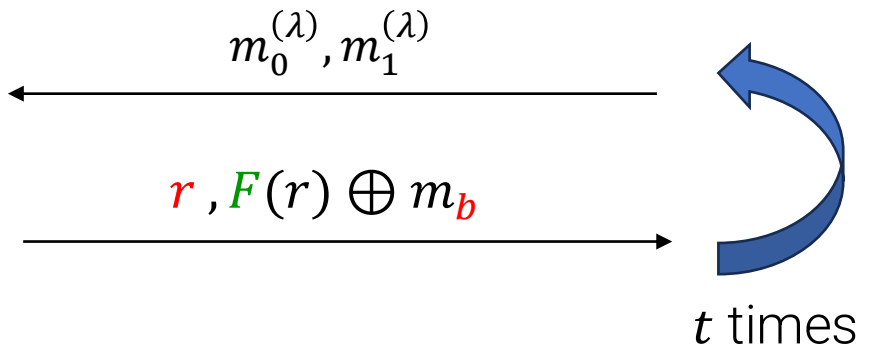
Win if $b' = b$

$\approx_c$
by PRF security
(reduction)



F

$b \leftarrow \{0,1\}$



b'

Win if $b' = b$

Building CPA-secure Encryption



$m_0^{(\lambda)}, m_1^{(\lambda)}$

r, s



t times



b'

Win if $b' = b$

$\approx_{\text{negl}}$

$b \leftarrow \{0,1\}$



$m_0^{(\lambda)}, m_1^{(\lambda)}$

$r, F(r) \oplus m_b$



t times



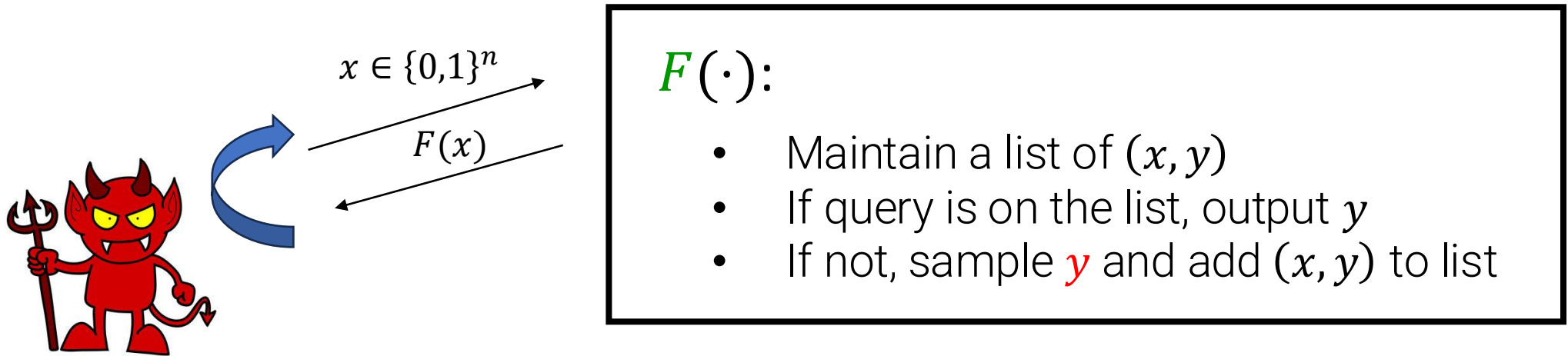
b'

Win if $b' = b$

F

$b \leftarrow \{0,1\}$

Pseudorandom Functions (PRFs)



If we repeatedly call $F(r_1), \dots, F(r_t)$ for independent $r_1, \dots, r_t$, we get independent $y_1, \dots, y_t$ unless some $r_i = r_j$.

Building CPA-secure Encryption



k

$b \leftarrow \{0,1\}$

$m_0^{(\lambda)}, m_1^{(\lambda)}$

r, s

(set $|r| = \lambda$ so that the t samples of r are distinct!)

t times



b'

Win if $b' = b$



F

$b \leftarrow \{0,1\}$

$m_0^{(\lambda)}, m_1^{(\lambda)}$

$r, F(r) \oplus m_b$

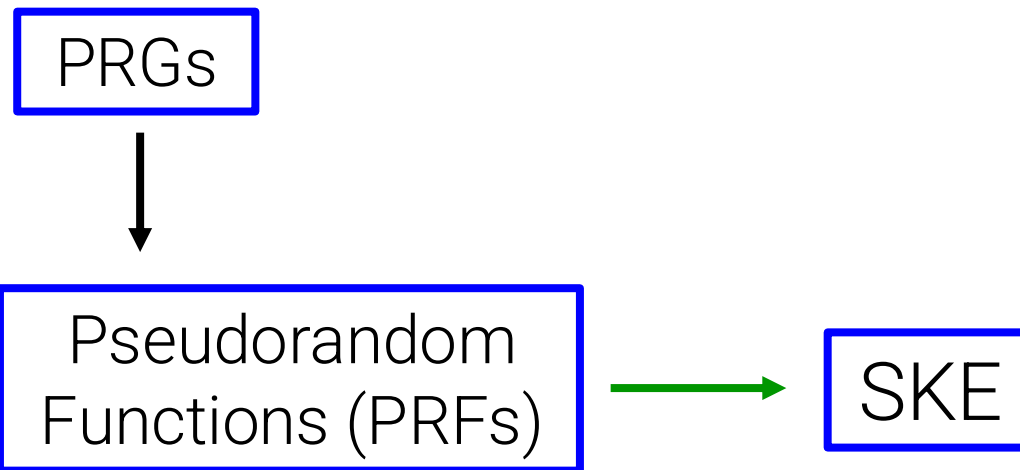
t times

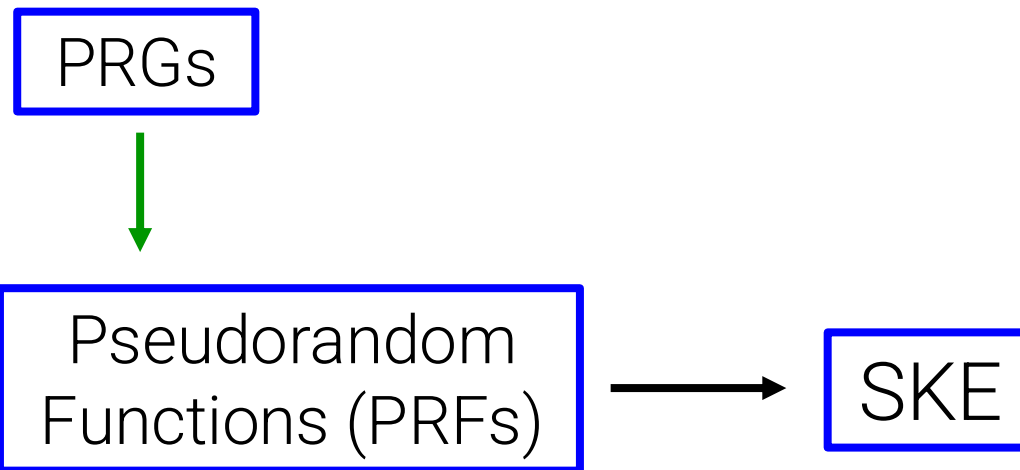


b'

Win if $b' = b$

$\approx_{\text{negl}}$





Building PRFs

Theorem (GGM86): if PRGs exist, then so do **PRFs!**

Goldreich

–

Goldwasser

–

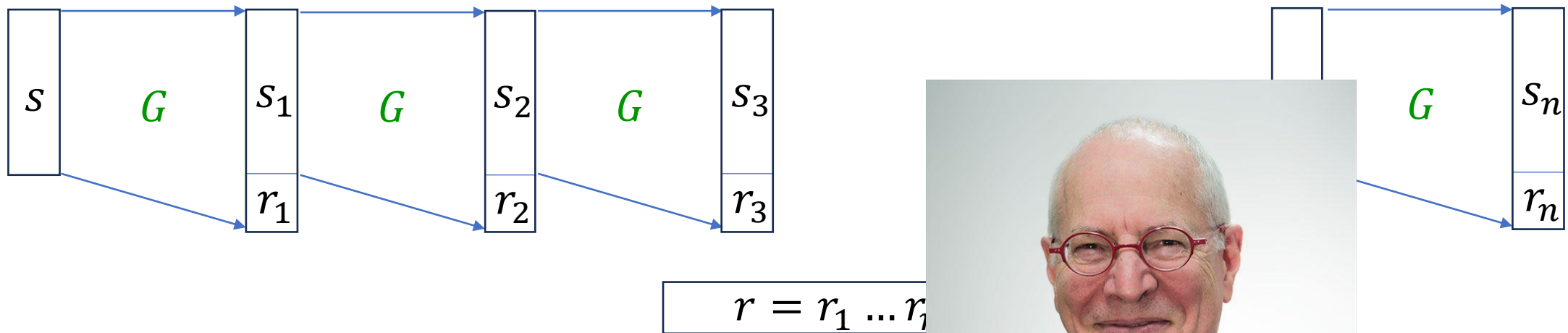
Micali



Building PRFs

Theorem (GGM86): if **PRGs** exist, then so do **PRFs**!

Proof: Length extension??

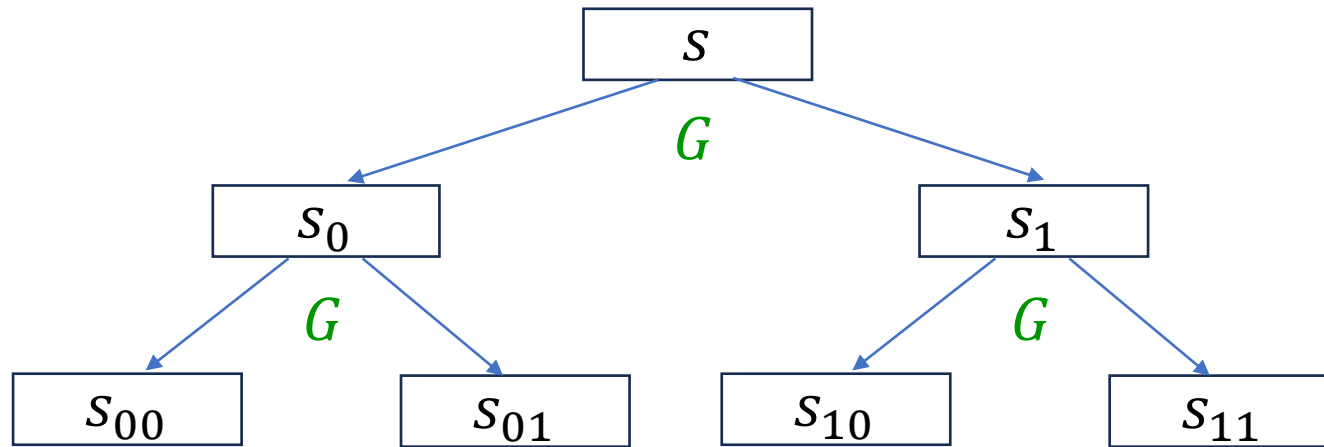


We are missing... a data structure!



Building PRFs

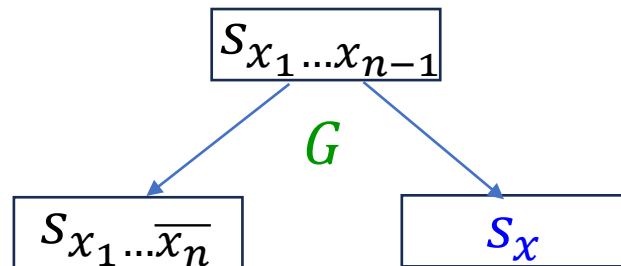
Let $G: \{0,1\}^\lambda \rightarrow \{0,1\}^{2\lambda}$ denote a **length doubling PRG**



Layer i : 2^i vertices

...

(suppose $x_n = 1$)



Define $\text{PRF}_s(x) = s_x$

Building PRFs

Let $G: \{0,1\}^\lambda \rightarrow \{0,1\}^{2\lambda}$ denote a **length doubling PRG**

Defining $G(s) = (G_0(s), G_1(s))$, this means:

$$\text{PRF}_s(x) = G_{x_n} \left(G_{x_{n-1}} \left(\dots \left(G_{x_1}(s) \right) \right) \right)$$

Even though tree has exp size, any leaf value $\text{PRF}_s(x)$
can be computed in polynomial time!

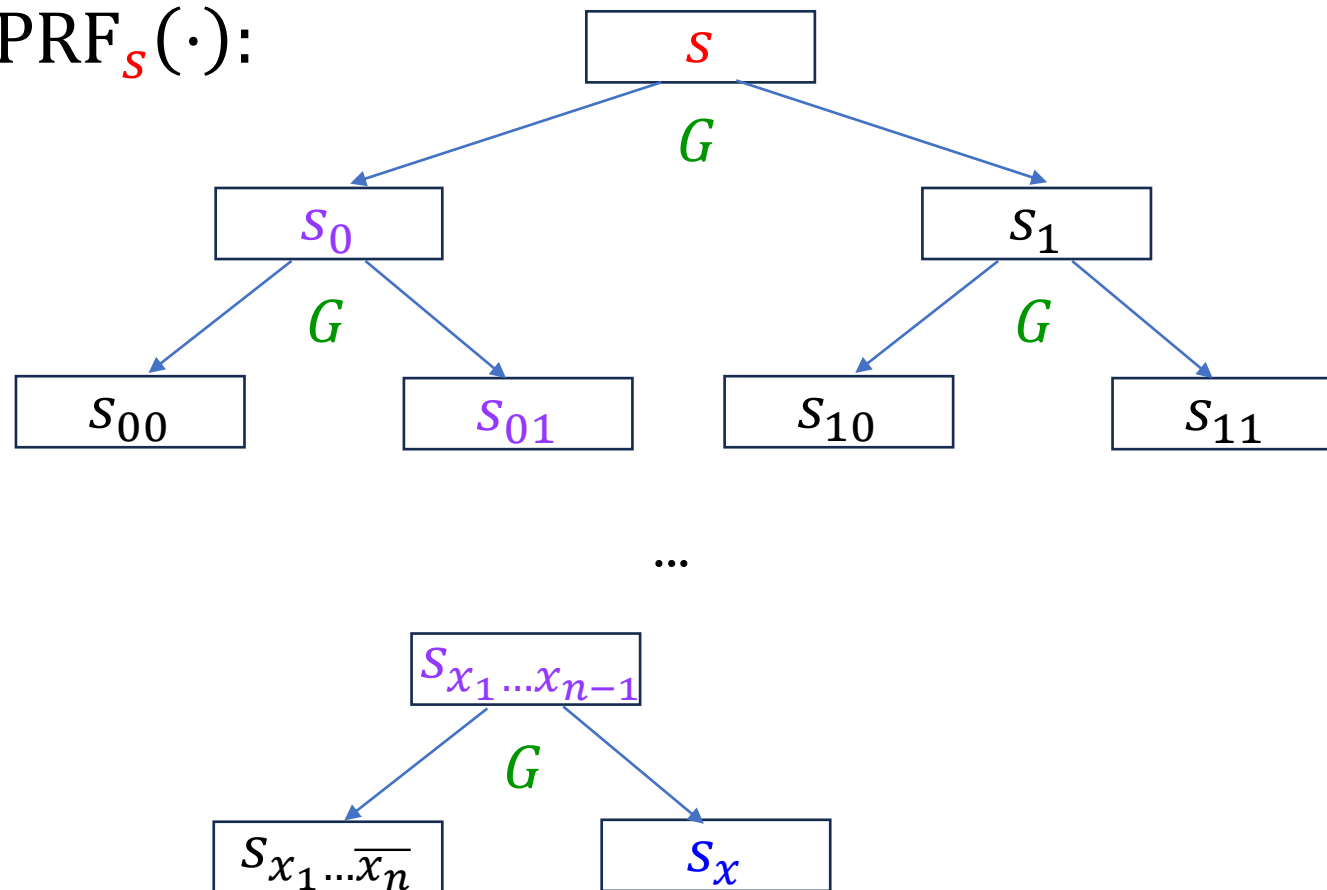


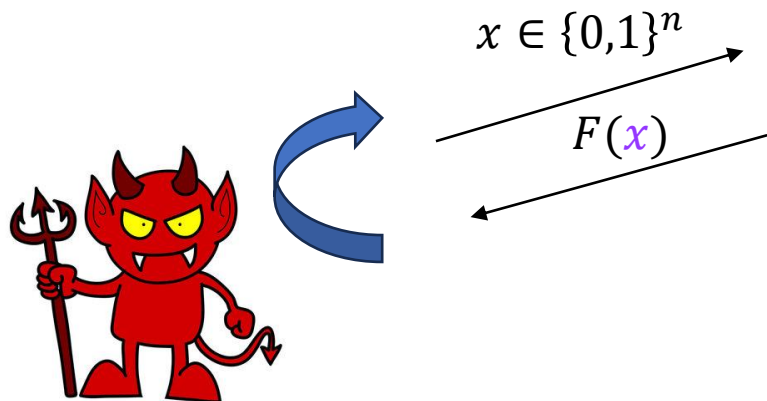
$x \in \{0,1\}^n$

$F(x)$

How do we prove
pseudorandomness?

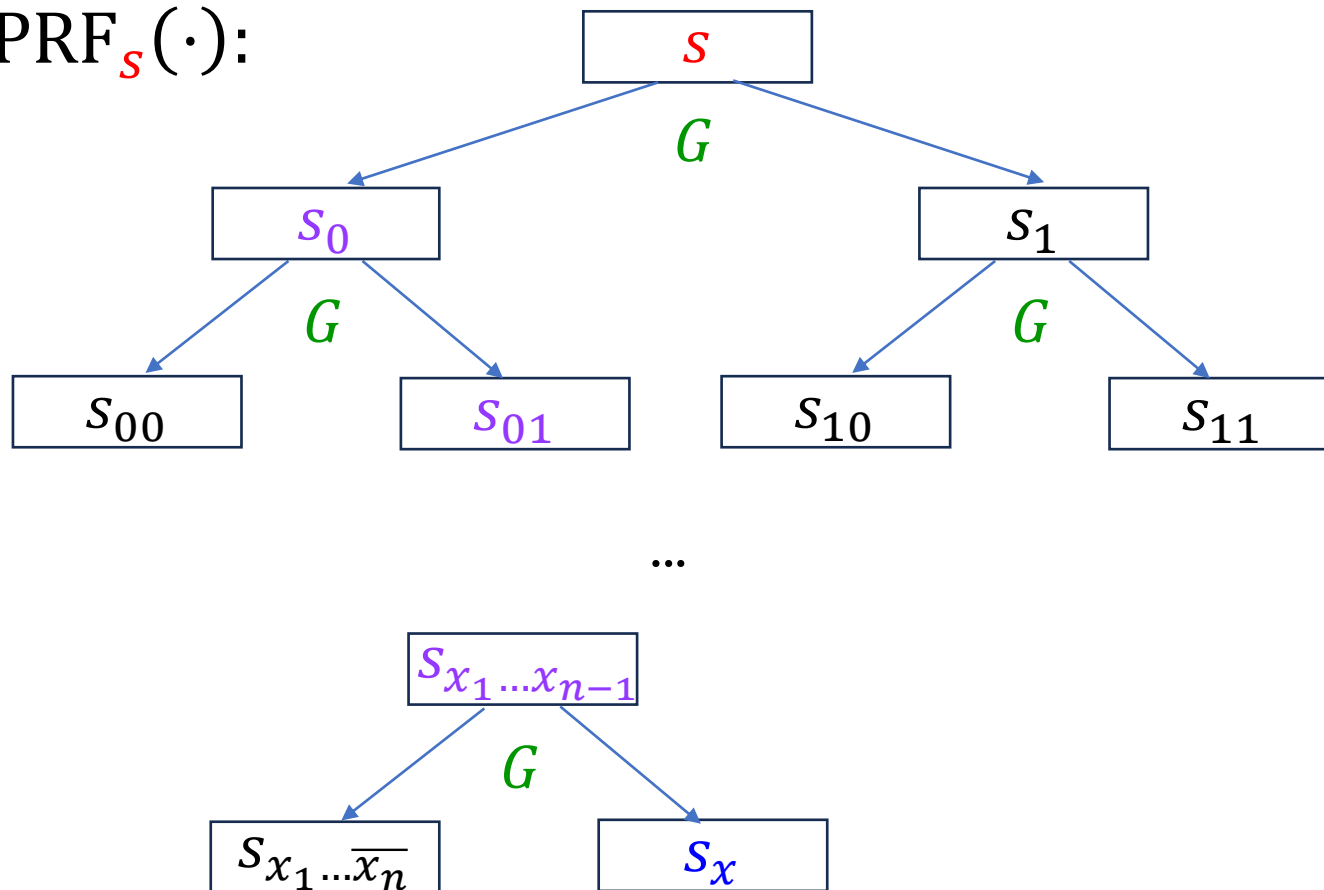
$\text{PRF}_s(\cdot)$:

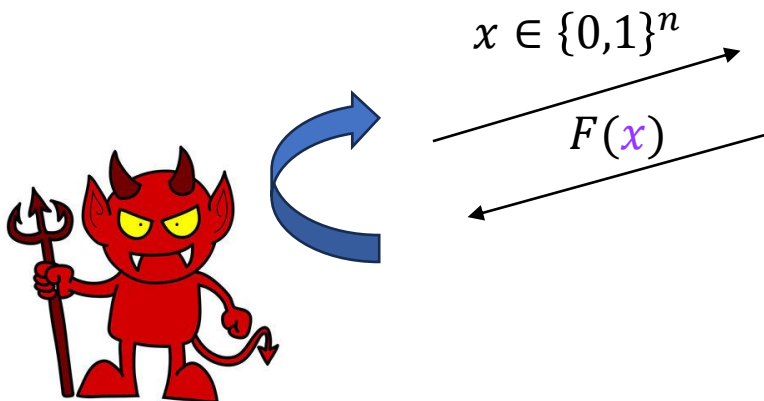




- $\text{PRF}_s(x)$ only accesses a poly-size path in the tree.

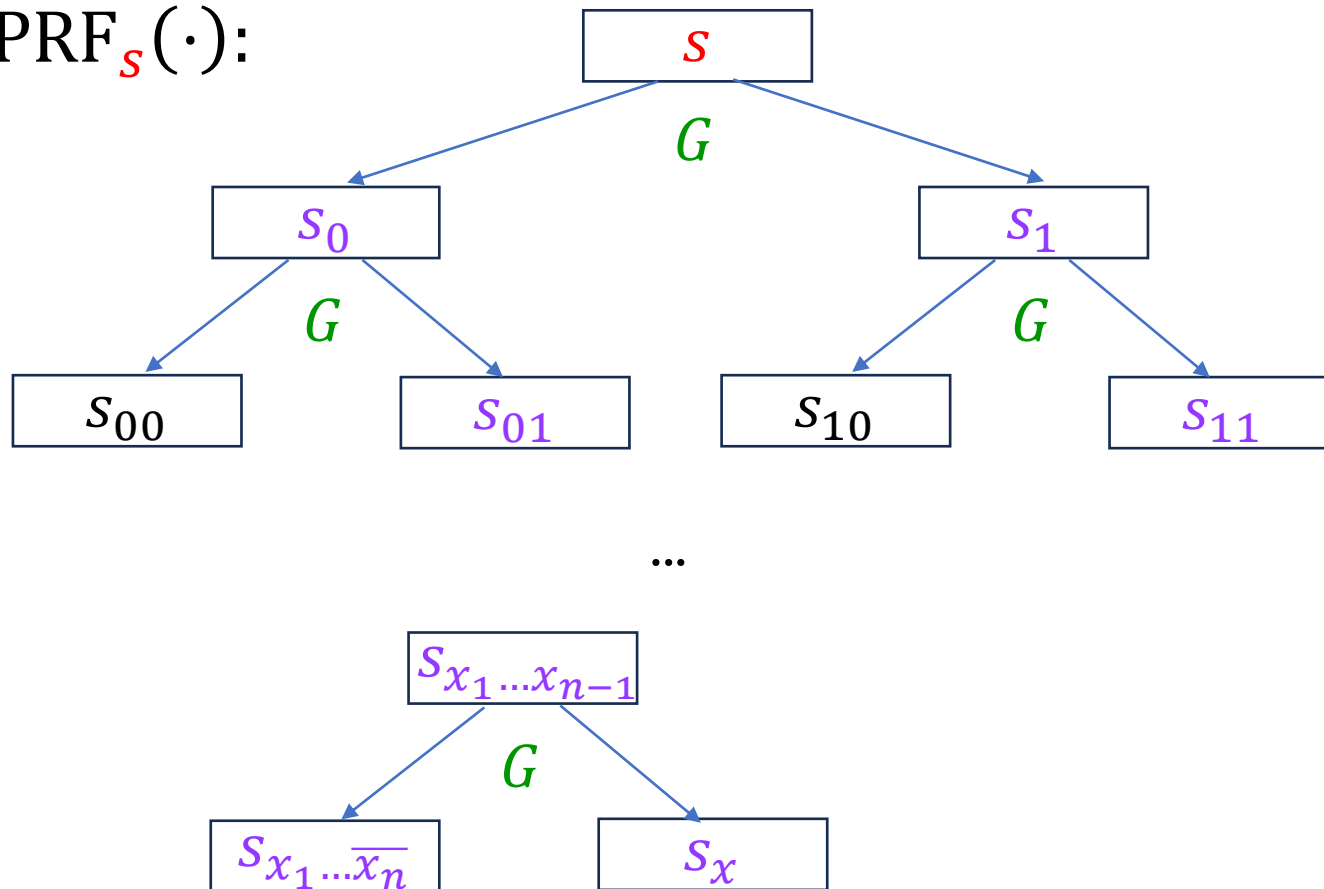
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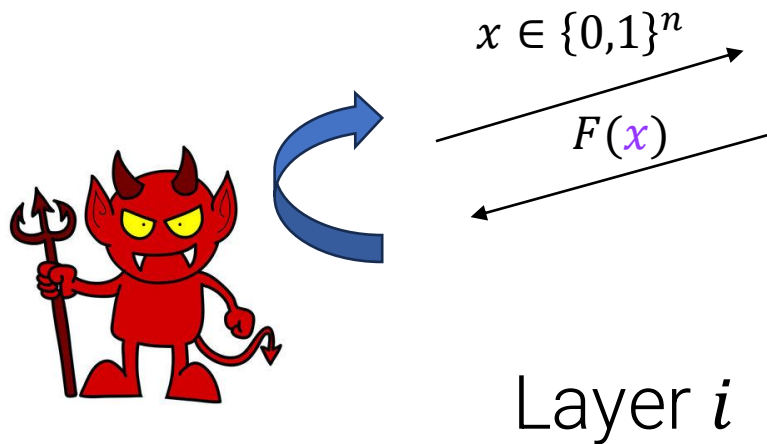




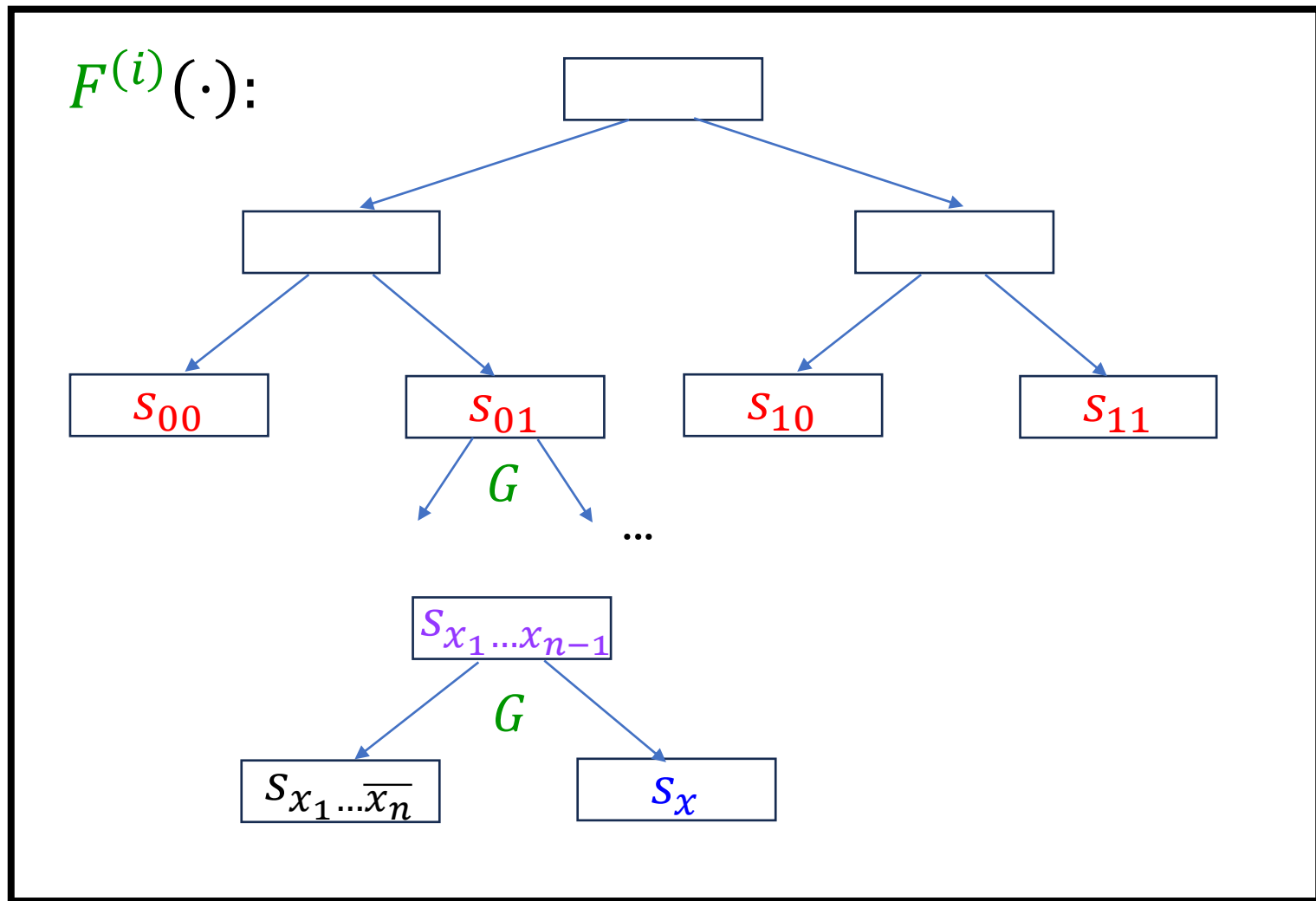
- $\text{PRF}_s(x)$ only accesses a poly-size path in the tree.
- t queries: union of t paths

$\text{PRF}_s(\cdot)$:





Function distribution $F^{(i)}$

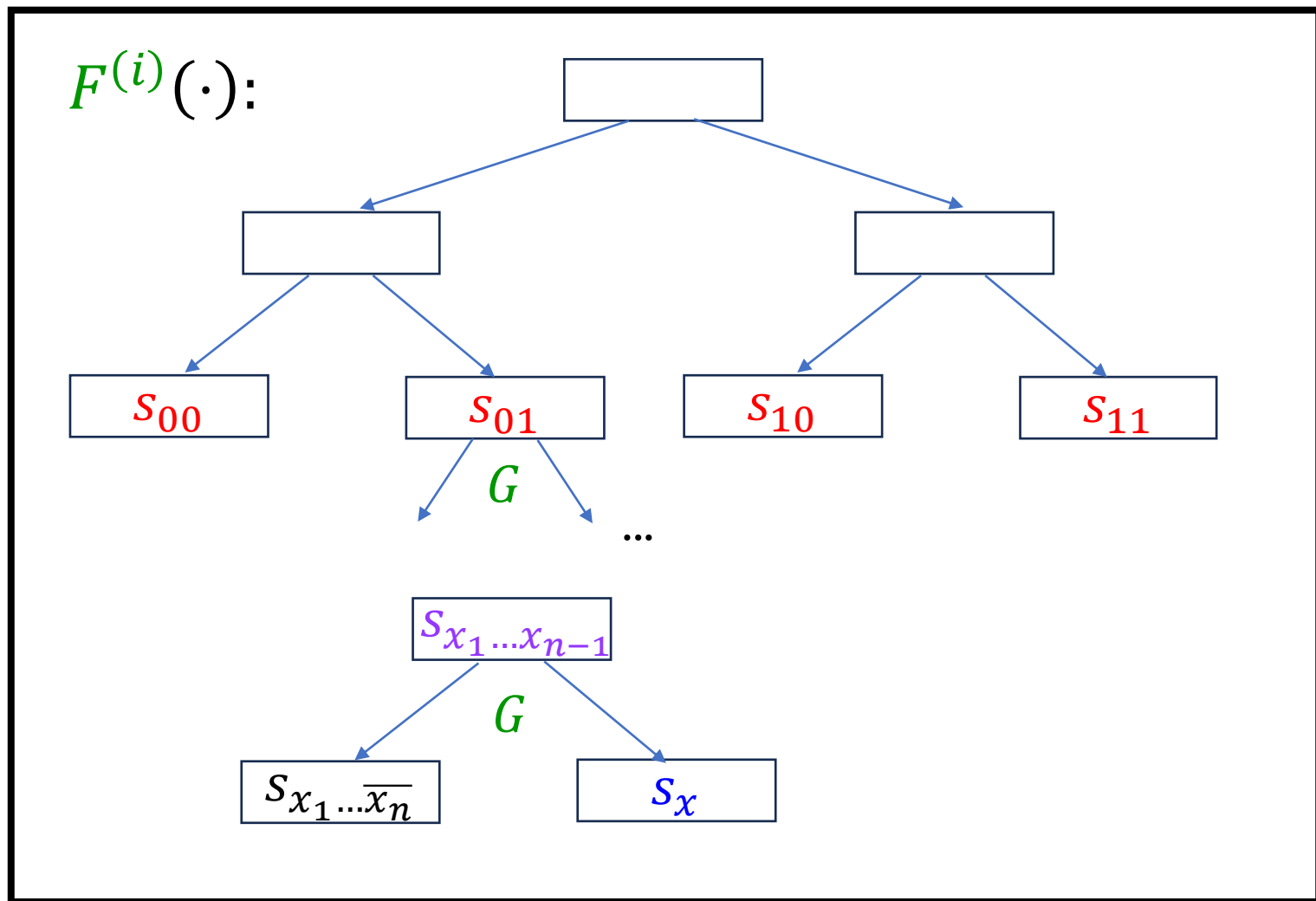


- $F^{(i)}(x)$ applies a uniformly random function to $x_1 \dots x_i \rightarrow s_{x_1 \dots x_i}$
- Computes s_x from $s_{x_1 \dots x_i}$ using G .


 $x \in \{0,1\}^n$
 $F(x)$

Layer i

Function distribution $F^{(i)}$



- $F^{(0)}(x) = \text{PRF}_s(x)$
- $F^{(n)}(x) = F(x)$ (random F)

Need:


 $F^{(i)}(\cdot)$
 $\approx_c$

 $F^{(i+1)}(\cdot)$

Need:  $F^{(i)}(\cdot)$ $\approx_c$  $F^{(i+1)}(\cdot)$

- We will use **another hybrid argument!**
- Worry: 2^i different seeds in $F^{(i)}$ -- can't change them one by one...
- It's ok: adversary only accesses poly many of them.



$x \in \{0,1\}^n$

$F(x)$

$F^{(i)}(\cdot)$:

- Maintain a list of $(x_1 \dots x_i, y)$
- If query is on the list, output $\text{PRF}_y(x_{i+1} \dots x_n)$
- If not, sample y and add (x, y) to list



$x \in \{0,1\}^n$

$F(x)$

$F^{(i+1)}(\cdot)$:

- Maintain a list of $(x_1 \dots x_i, y_0, y_1)$
- If query is on the list, output $\text{PRF}_{y_{x_{i+1}}}(x_{i+2} \dots x_n)$
- If not, sample y_0, y_1 and add (x, y_0, y_1) to list



$x \in \{0,1\}^n$

$F(x)$

$F^{(i)}(\cdot)$:

- Maintain a list of $(x_1 \dots x_i, y_0, y_1)$
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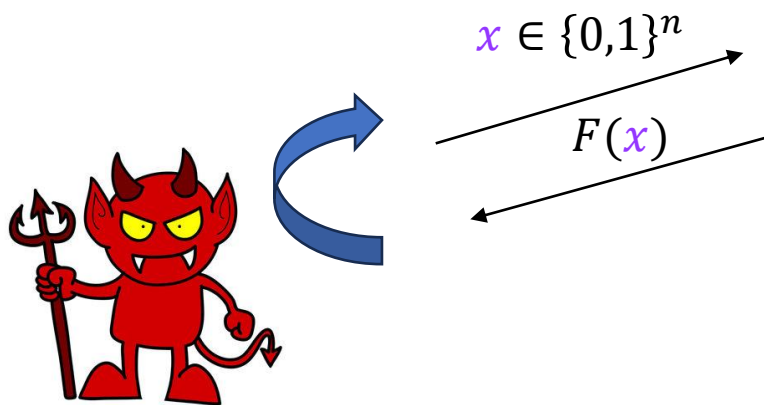


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$F(x)$

$F^{(i+1)}(\cdot)$:

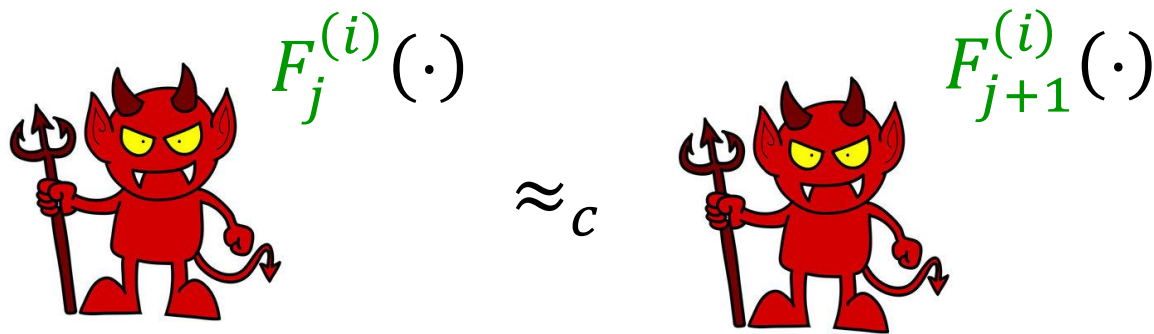
- Maintain a list of $(x_1 \dots x_i, y_0, y_1)$
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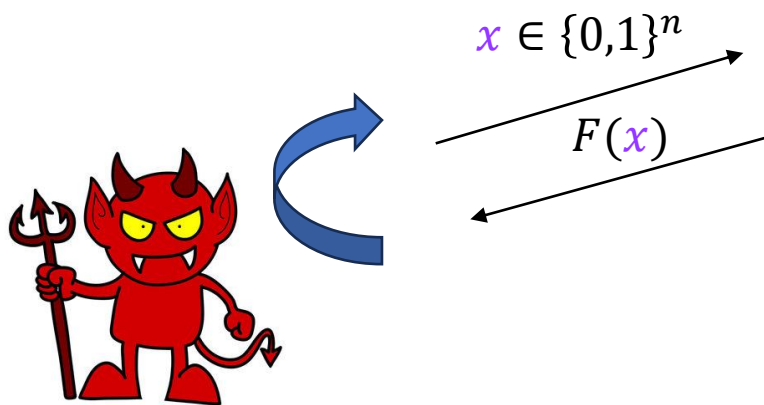


$F_j^{(i)}(\cdot)$:

- Maintain a list of $(x_1 \dots x_i, y_0, y_1)$
- If query is on the list, output $\text{PRF}_{y_{x_{i+1}}}(x_{i+2} \dots x_n)$
- If not:
 - (First j times) sample y and add $(x, G(y))$
 - (Otherwise) sample y_0, y_1 and add (x, y_0, y_1)

PRG security implies:

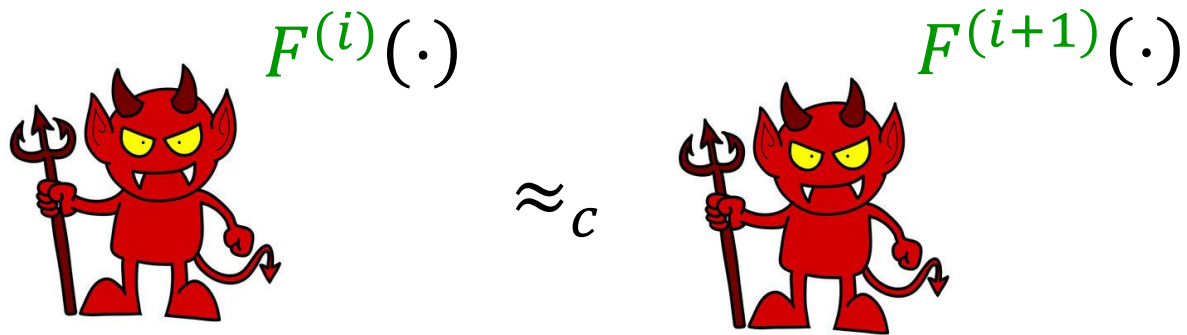


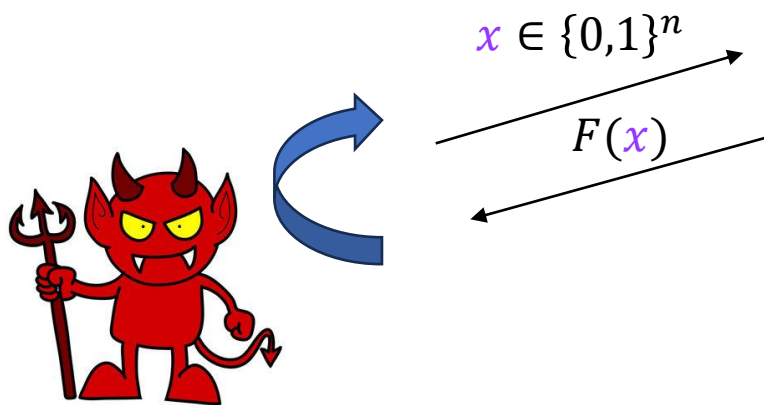


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Therefore:

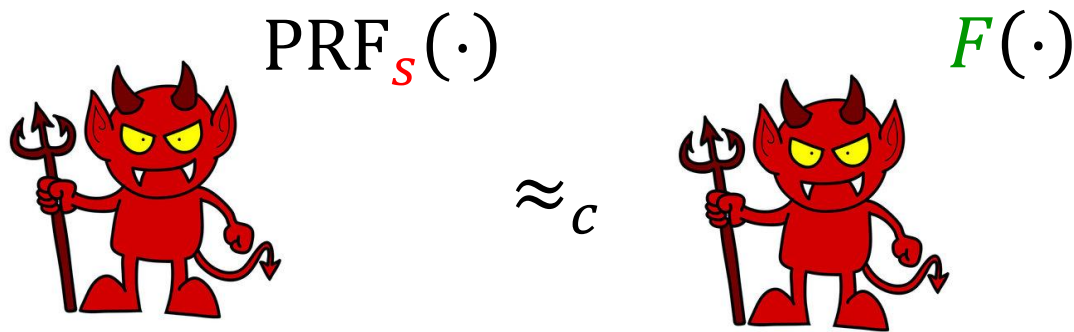




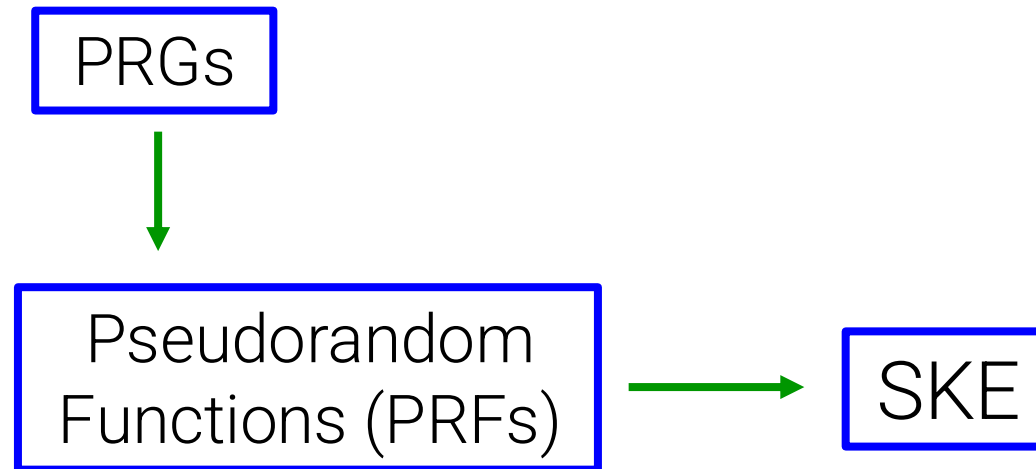
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 - (First j times) sample y and add $(x, G(y))$
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Finally:



Plan for this week



Next: what do “practical” SKE schemes look like?